



VIT[®]

Vellore Institute of Technology

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VELLORE INSTITUTE OF TECHNOLOGY (VIT), CHENNAI

School of Electrical Engineering (SELECT)

Advanced Cloud Computing Project Report

FindItVITC

A Cloud-Native Serverless Lost & Found System and Emergency Broadcast
Network Powered by AWS



Course Code & Title: BECE355L - Advanced Cloud Computing

Team Members & Reg Nos: E Jerish - 24BEE1102

Project Repository & Live URL:
GitHub Repo: <https://github.com/finditvitc/finditvitc.github.io>
Live URL: <https://finditvitc.github.io>
AWS Region: ap-south-1 (Mumbai)

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1. Problem Definition, Motivation

1.1 The Real-World Campus Challenge

On large residential higher-education institutions such as Vellore Institute of Technology (VIT) Chennai, thousands of students, faculty members, and visitors navigate a vast physical campus comprising multiple Academic Blocks (AB1 through AB5), central libraries, sprawling multi-sport complexes, residential hostels, dining facilities, and auditoriums on a daily basis. In such an expansive and densely populated environment, personal belongings—including high-value electronics (laptops, smartphones, wireless earbuds, calculators), official identification credentials (student identity cards, driver licenses), keys, backpacks, and academic materials—are routinely misplaced.

Historically, the lost-and-found recovery process on campus has been severely fragmented and manual. When items are misplaced, custody is scattered across disconnected physical security desks, departmental notice boards, block entrances, and informal social media chat groups. Students must physically walk between multiple offices over several days to inquire whether their item was turned in. Staff members maintain informal paper logbooks with vague, subjective textual descriptions (e.g., 'blue bottle' or 'black laptop bag') that lack standard categorization. As a result, the vast majority of misplaced campus belongings go unrecovered, creating unnecessary financial distress for students and administrative overhead for campus security.

Simultaneously, the campus lacks a centralized, real-time mechanism to broadcast high-priority safety notices and emergency instructions directly to students' active web sessions. Traditional bulk emails often experience delivery latency or go unread during time-critical incidents such as severe weather hazards, localized facility closures, or security drills.

1.2 Motivation & Architectural Solution

The motivation behind FindItVITC is to unify campus lost-and-found recovery and emergency safety communications into a single, high-performance, cloud-native web platform tailored specifically for the VIT Chennai community. By transitioning from WhatsApp groups to a centralized, serverless digital grid, every reported item is cataloged with structured metadata (22 campus zones, 11 school departments, precise timestamps, and status lifecycle flags).

Crucially, the platform introduces AI-assisted matching powered by Amazon Rekognition computer vision. Rather than relying on subjective human descriptions, the system automatically analyzes uploaded item photographs to detect visual objects, hierarchical classifications, and dominant subject colors. A multi-factor similarity algorithm continuously evaluates lost and found reports to proactively recommend high-confidence candidate matches. This automated intelligence meaningfully improves recovery rates, minimizes search friction from days to seconds, and provides administrators with instantaneous emergency broadcast and stand-down capabilities.

2. AWS Services Used

FindItVITC is engineered strictly following modern cloud-native serverless architecture principles. The entire backend is hosted on Amazon Web Services (AWS) in the Asia Pacific (Mumbai) region (ap-south-1). Every service was selected to eliminate idle server maintenance costs (\$0/month under the AWS Free Tier), guarantee automatic horizontal scaling, and ensure high availability across multiple Availability Zones.

2.1 Amazon Cognito — Identity, Domain-Restricted Auth & Role RBAC

What It Is: Amazon Cognito is a fully managed customer identity and access management (CIAM) service that provides scalable user directories, self-service registration, authentication, multi-factor verification, and role-based access control (RBAC).

Why Chosen Over Alternatives: Building a custom authentication system requires managing password hashing (bcrypt/Argon2), session token rotation, salt generation, and email verification servers, which introduces substantial security liability. Self-hosted auth alternatives (e.g., Auth0 or custom OAuth servers) either incur monthly subscription fees or require continuous EC2 server maintenance. Amazon Cognito delivers an enterprise-grade, OAuth2/OIDC-compliant identity pool with 50,000 free monthly active users (MAUs), seamless JWT token integration with Amazon API Gateway, and native AWS Lambda pre-signup execution triggers.

How Used in FindItVITC: In FindItVITC, Cognito User Pool (ap-south-1_K5c6MntVx, Client ID: 5068gn9iktIj9670vndtdn125m) manages the complete user lifecycle. A dedicated Pre Sign-up Lambda Trigger (pre_signup.py) enforces a strict domain validation rule: self-registration is restricted exclusively to official '@vitstudent.ac.in' email addresses, rejecting external domains. Cognito dispatches secure 6-digit confirmation codes for email verification. Furthermore, Cognito User Groups ('Admin' and 'Security') enforce Role-Based Access Control (RBAC): only verified security personnel and administrators can access the emergency broadcast console and terminate active campus alerts.

2.2 Amazon S3 — High-Durability Object Storage & Event Trigger Source

What It Is: Amazon Simple Storage Service (S3) is an industry-leading object storage service providing 99.999999999% (11 9s) of data durability, limitless scalability, fine-grained access control, and native event notifications.

Why Chosen Over Alternatives: Storing image binaries inside NoSQL document databases (e.g., base64 in DynamoDB) causes severe database bloat, exceeds DynamoDB's 400KB item size limit, and degrades query latency. Similarly, streaming multi-megabyte image uploads through API Gateway and Lambda rapidly exhausts Lambda memory allocations and hits API Gateway's 10MB payload ceiling. Amazon S3 enables direct-to-bucket uploads via presigned URLs, offloading all network bandwidth and binary transfer overhead from the serverless compute layer.

How Used in FindItVITC: In FindItVITC, a private S3 bucket (FindItVITC-photos-694442891642-ap-south-1) with Block Public Access enabled stores all user-submitted item photographs. When a student attaches a photo, the frontend requests a short-lived, cryptographically signed presigned PUT URL from

the backend Uploads Lambda. The browser streams the compressed JPEG binary directly to S3 over HTTPS. S3 emits s3:ObjectCreated events to trigger automated Rekognition analysis, and the Items Lambda generates 7-day presigned GET URLs for authorized frontend rendering.

2.3 Amazon Rekognition — Deep Learning Computer Vision & Auto-Tagging

What It Is: Amazon Rekognition is a fully managed, deep learning-based computer vision service that analyzes images and video to identify objects, labels, scenes, visual concepts, text, and dominant color properties without requiring custom ML model training.

Why Chosen Over Alternatives: Deploying and fine-tuning custom convolutional neural networks (CNNs or YOLO models) requires provisioning expensive GPU-enabled virtual machines (e.g., AWS EC2 G4dn/P3 instances) with high baseline monthly costs. Open-source local CV models running on serverless CPU Lambdas suffer from cold-start latency and strict package size constraints. Amazon Rekognition provides pre-trained, highly accurate visual taxonomies accessible via simple API calls with zero infrastructure management and millisecond execution times.

How Used in FindItVITC: In FindItVITC, the Rekognition DetectLabels API processes every uploaded item photograph. The vision pipeline uses an adaptive multi-confidence evaluation (45% -> 30%) with ImageProperties feature extraction. It automatically extracts top object labels (e.g., 'Laptop', 'Headphones', 'Backpack', 'Identity Card'), parent categories ('Electronics', 'Luggage'), and chromatic dominant colors ('Blue', 'Pink', 'Black'). These visual tags are stored alongside the item record in DynamoDB (converted to Decimal types to ensure Boto3 serialization compatibility) and directly fuel the automated AI similarity engine.

2.4 AWS Lambda — Serverless Compute & Event-Driven Microservices

What It Is: AWS Lambda is a serverless, event-driven compute service that executes backend code in isolated execution environments in response to HTTP requests or cloud events, automatically managing compute provisioning, patching, and horizontal scaling.

Why Chosen Over Alternatives: Traditional monolithic web servers hosted on Amazon EC2 or virtual private servers require 24/7 provisioning, operating system maintenance, load balancer configuration, and continuous baseline expenditures even during zero-traffic periods (such as campus holidays and night hours). AWS Lambda executes compute on-demand with 1ms billing granularity, automatically scales to hundreds of concurrent student requests during peak hours, and incurs exactly \$0.00 in idle compute cost.

How Used in FindItVITC: In FindItVITC, four Python 3.12 microfunctions power the entire business logic in ap-south-1:

- **ItemsFunction:** Handles CRUD operations, query filtering across 22 campus zones and 9 categories, status transitions (Open/Claimed/Resolved), and multi-factor matching algorithms.
- **RekognitionTriggerFunction:** Handles S3 image creation events and executes synchronous /rekognition/analyze labeling.

- AlertsFunction: Authenticates admin permissions, publishes broadcasts to SNS, dispatches emails via Amazon SES, logs alerts to DynamoDB, and executes stand-down terminations.
- UploadPresignFunction: Generates secure, short-lived presigned PUT URLs for direct browser-to-S3 uploads.
- PreSignUpTrigger: Intercepts Cognito registration requests to enforce the @vitstudent.ac.in domain validation.

2.5 Amazon DynamoDB — Low-Latency NoSQL Document Database

What It Is: Amazon DynamoDB is a fully managed, serverless, key-value and document NoSQL database designed to deliver single-digit millisecond response times at any scale with built-in security, continuous backups, and multi-region replication.

Why Chosen Over Alternatives: Relational database engines (Amazon RDS PostgreSQL / MySQL) require persistent database instances, connection pool management, and fixed monthly provisioning costs regardless of traffic. Relational schemas are also rigid when storing dynamic computer vision tags and varying item attributes. DynamoDB offers schema flexibility, native JSON document support, On-Demand (PAY_PER_REQUEST) capacity billing with zero idle cost, and single-digit millisecond reads and writes.

How Used in FindItVITC: In FindItVITC, two dedicated DynamoDB tables manage application state:

1. FindItVITC-Items: Primary key 'id' (UUID string). Configured with a Global Secondary Index (GSI) named 'TypeCreatedAtIndex' on partition key 'type' ('lost'/'found') and sort key 'createdAt' (ISO-8601 timestamp), enabling sub-second feed pagination and filtering without full table scans.
2. FindItVITC-Alerts: Primary key 'id' (UUID string). Stores emergency broadcast history, severity levels, target zones, active status flags, resolution timestamps, and idempotency keys.

2.6 Amazon SNS & Amazon SES — Multi-Channel Emergency Broadcast Hub

What It Is: Amazon Simple Notification Service (SNS) is a fully managed pub/sub messaging service for high-throughput, fan-out delivery. Amazon Simple Email Service (SES) is a cost-effective, cloud-based email service for sending transactional and bulk notifications.

Why Chosen Over Alternatives: Managing dedicated SMTP mail servers or building custom WebSocket notification clusters requires continuous server maintenance, IP reputation warming, and connection state management. Amazon SNS and SES offload all message queueing, subscriber registry, and delivery infrastructure, ensuring campus safety alerts reach student inboxes and notification banners within seconds.

How Used in FindItVITC: In FindItVITC, the Alerts Lambda publishes broadcasts to SNS topic 'FindItVITC-EmergencyAlerts' (ARN: arn:aws:sns:ap-south-1:694442891642:FindItVITC-EmergencyAlerts) with message attributes for severity filtering. Simultaneously, direct SES email dispatch transmits formatted emergency bulletins to registered campus administrators and students, while the frontend continuously syncs the live header banner across all active student browser tabs.

2.7 Amazon API Gateway — REST API Management & Security Layer

What It Is: Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure REST and WebSocket APIs at any scale.

Why Chosen Over Alternatives: API Gateway provides native integration with Cognito User Pool Authorizers, built-in SSL/TLS termination, request validation, automated DDoS protection via AWS Shield, and customizable CORS handling without requiring custom NGINX reverse proxies.

How Used in FindItVITC: In FindItVITC, a regional REST API Gateway (Stage: prod, URL: <https://8d3aayrch4.execute-api.ap-south-1.amazonaws.com/prod>) routes client HTTP requests to backend Lambdas. Endpoints (/items, /alerts, /uploads/presign, /rekognition/analyze) are protected by Cognito authorizer tokens. Custom GatewayResponses ('DEFAULT_4XX', 'DEFAULT_5XX') are configured to inject Access-Control-Allow-Origin: '*' headers on all auth errors, preventing browser CORS failures.

3. Modules and Description

FindItVITC is structured into six cohesive functional modules, each addressing a specific stage of the campus lost-and-found lifecycle and public safety operations.

Module Name	Core Purpose & Capabilities	AWS Services & Frontend Components
Authentication & Registration	Enforces @vitstudent.ac.in domain validation, 6-digit email confirmation, and Admin/Security RBAC verification.	Amazon Cognito, Pre Sign-up Lambda, AuthContext.jsx, LoginPage.jsx, AuthModal.jsx
Report Lost / Found Item	Captures item reports with image compression, direct S3 upload, instant Oms preview, and Rekognition auto-tagging across 22 campus zones.	Amazon S3, UploadPresign Lambda, Rekognition Lambda, ReportItemPage.jsx, imageFallbacks.js
AI-Assisted Matching Engine	Calculates statistical similarity between lost & found reports using a 4-factor scoring algorithm (75% confidence threshold) with 1-click email claims.	Amazon Rekognition, Items Lambda (matching_engine.py), DynamoDB, MatchSuggestionsModal.jsx, ItemDetailPage.jsx
Item Feed & Search/Filter	Real-time search across titles, descriptions, and AI tags with multi-parameter filtering (Type, 9 Categories, 22 Zones, Status) and responsive dark/light UI.	Amazon DynamoDB (TypeCreatedAtIndex GSI), Items Lambda, FeedPage.jsx, ItemCard.jsx, Navbar.jsx
Emergency Alert & Stand-Down	Gated console for administrators to broadcast multi-severity safety alerts via SES email and in-app banners, with 1-click termination stand-down.	Amazon SNS, Amazon SES, Alerts Lambda (alerts.py), DynamoDB (FindItVITC-Alerts), AdminAlertPanel.jsx, EmergencyBanner.jsx
My Reports / User Dashboard	Personalized management dashboard for students to track active reports, view candidate matches, update statuses (Claimed/Resolved), or delete entries.	Amazon DynamoDB (userId index), Items Lambda (ownership IDOR verification), MyReportsPage.jsx

3.1 Detailed Module Descriptions

Module 1: Authentication & Registration

This module secures the application by verifying campus identities. Students sign up using their official VIT email address. The Cognito Pre Sign-up Lambda trigger intercepts registration requests and verifies that the email ends with '@vitstudent.ac.in'. Accounts from external domains (e.g. Gmail, Yahoo) are immediately rejected at the server level. Upon registration, Cognito dispatches a 6-digit confirmation code. Authenticated sessions are stored securely in sessionStorage (findit_session_user), preventing unauthorized access while ensuring seamless multi-tab security.

Module 2: Report Lost/Found Item

Students and staff submit reports through an intuitive intake form. When an image is attached, the client canvas compresses the image to ~25KB JPEG data URL for instant 0ms preview. The client requests a presigned PUT URL from UploadPresignFunction and streams the binary directly to Amazon S3. Amazon Rekognition analyzes the image to extract visual tags and dominant colors. The user selects from 22 verified VIT Chennai campus locations (AB1–AB5, Library, Gazebo, Sports complex, etc.) and 11 academic schools (SCOPE, SENSE, SELECT, etc.).

Module 3: AI-Assisted Matching Engine

The matching engine solves the core retrieval problem by cross-referencing opposing item types (Lost vs. Found). The algorithm computes a composite 4-factor weighted similarity score:

- Category Match (30%): Direct match between item categories.
- Rekognition Visual Labels (35%): Jaccard similarity index measuring overlap between computer vision tags.
- Location Zone Proximity (20%): Proximity match across the 22 campus locations.
- Keyword & Description Overlap (15%): Natural language token overlap.

Candidate matches with similarity $\geq 75\%$ are highlighted with detailed reasoning, and students can initiate 1-click email claim coordination via pre-filled match templates.

Module 4: Item Feed & Search/Filter

The public item feed provides sub-second search and multi-parameter filtering across all campus reports. Powered by DynamoDB's Global Secondary Index (TypeCreatedAtIndex), items are retrieved in reverse chronological order. Students can filter by Type (Lost/Found), 9 Categories, 22 Campus Locations, and lifecycle status. The UI features responsive Tailwind CSS v3 styling with high-contrast Dark/Light mode support.

Module 5: Emergency Alert & Broadcast Module (Admin Only)

Restricted exclusively to members of the Cognito 'Admin' and 'Security' user pool groups, this console allows public safety personnel to compose and publish campus-wide emergency notices. Alerts are categorized by urgency (Critical Emergency, Safety Warning, Security Incident, Campus Notice) and broadcast via Amazon SNS and direct Amazon SES email. The global EmergencyBanner component pins active alerts to the top of all user screens. An integrated Stand-Down Termination feature enables administrators to deactivate active alerts with 1 click across the entire campus grid.

Module 6: My Reports / User Dashboard

The dashboard provides students with centralized custody management over their submissions.

Students can track report statuses, review AI candidate matches, update statuses (Mark as Claimed / Handed Over / Re-open), and delete resolved records. All update requests enforce IDOR (Insecure Direct Object Reference) ownership checks in Lambda, ensuring users can only modify their own submissions unless holding administrator privileges.

4. Implementation Details

This section presents key architectural screens from the live production deployment (<https://finditvitc.github.io>) alongside representative code listings extracted directly from the project repository.

4.1 Key Screen Captures

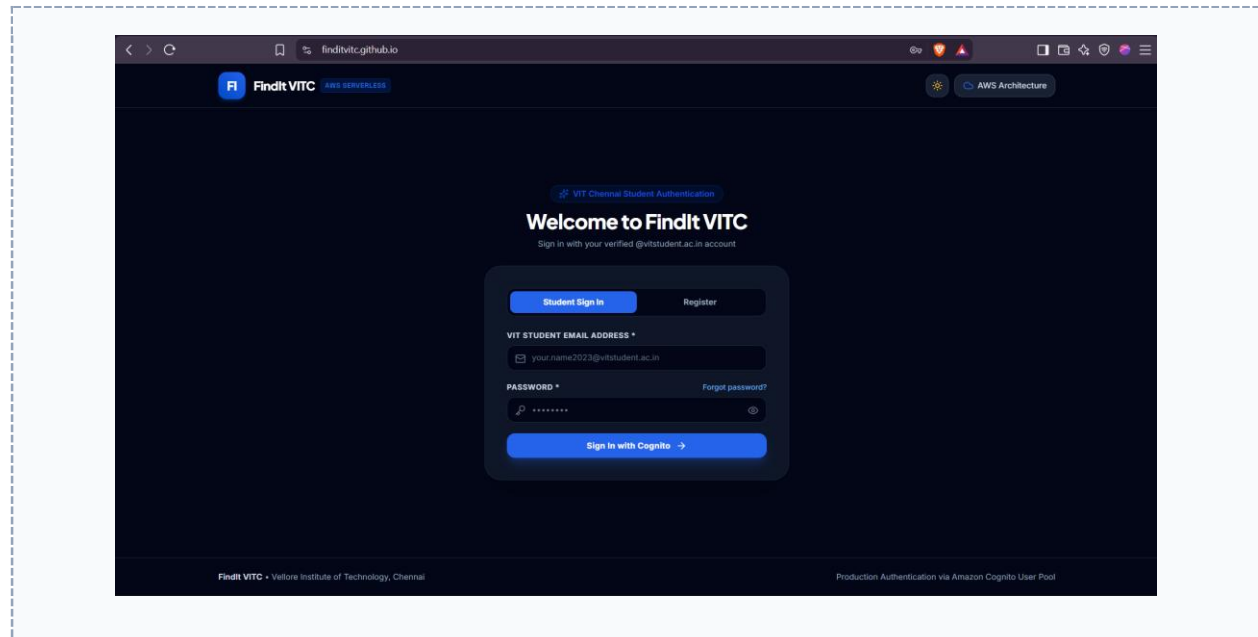


Figure 1: Student Authentication Portal & Domain Validation — Demonstrates the login and registration interface enforcing @vitstudent.ac.in domain validation with 6-digit confirmation code verification.

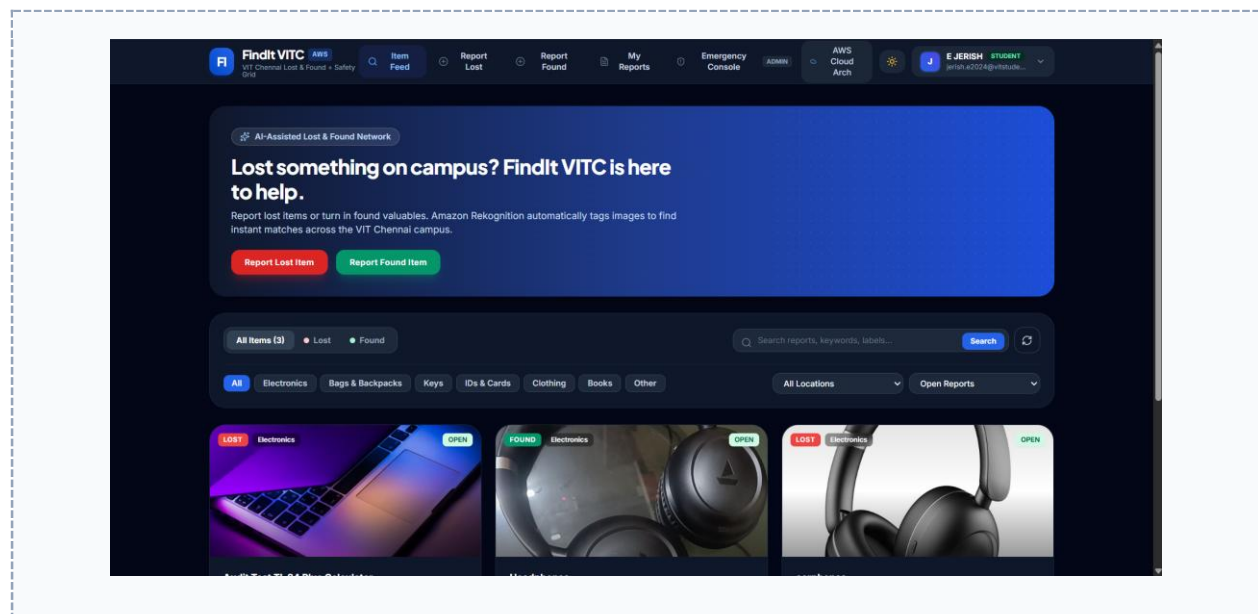


Figure 2: Live Campus Lost & Found Feed — Shows the responsive feed with multi-parameter filtering by Type (Lost/Found), 9 Categories, and 22 VIT Chennai campus locations in dark/light mode.

Report a Lost Item

Fill in the details below. Take a photo or upload an image for Amazon Rekognition AI auto-matching. Lost Found

ITEM TITLE / NAME * 0/100
Geometry box

CATEGORY * Electronics DATE & TIME LAST SEEN * 20-09-2026 12:40
Cannot be in the future.

CAMPUS LOCATION * AB3
Or specify exact room/cara (e.g. '3rd floor study carrel #14')

DETAILED DESCRIPTION & DISTINGUISHING FEATURES 0/1000
Mention distinctive stickers, scratches, colors, contents, brand names, or specific markings...

ITEM PHOTO & AMAZON REKOGNITION VISION AUTO-TAGGING
Take a photo with your camera or upload an image to auto-detect labels and colors. Auto-Extract Tags

 Auto-Detected AI Labels (used for matching):
Person x Brush x Device x Adult x
Red x
 Add custom tag (e.g. 'Blue', 'Sticker') Add Tag

Retake Change

YOUR CONTACT EMAIL / PHONE *
jerish.e2024@vltstudent.ac.in

Figure 3: Report Item Form with Amazon Rekognition Auto-Tagging — Displays real-time image upload, canvas compression, and automatic computer vision label extraction (e.g. Laptop, Electronics, Space Gray).

FindIt VITC Item Feed Report Lost Report Found My Reports Emergency Console ADMIN AWS Cloud Arch E JERISH STUDENT jerish.e2024@vltstude...

My Campus Reports & AI Match Finder 6 active + File New Report

Tracking reports filed by **E JERISH** (jerish.e2024@vltstudent.ac.in). FindIt VITC scans Amazon Rekognition features for matching items automatically.

SELECT A REPORT TO VIEW MATCHES

- LOST** Electronics **Geometry box** AB3 OPEN
- FOUND** Electronics **geometry box** AB2 OPEN
- LOST** Electronics **gb** AB1 OPEN
- LOST** Electronics **Geometry Box** AB2 OPEN
- LOST** Electronics **Audit Test TI-84 Plus Calculator** AB1 OPEN
- LOST** Electronics **earphones** OPEN

FOUND geometry box AB2 Status: Mark Claimed Mark Resolved

AI-Suggested Opposing Matches 5 matches found
Scored using Amazon Rekognition tags, Category, Location zone, and Keyword similarity

LOST Geometry Box AI MATCH CONFIDENCE 71.1% POTENTIAL MATCH
No additional details provided.
AB2 • Reported: 9/20/2026 Contact / Claim

Match Rationale:

- Exact category match: Electronics
- Matching visual tags: [person]
- Same location: AB2
- Title keywords match: [geometry, box]

Category	Vision Tags	Location	Keywords
100%	17.3%	100%	100%

LOST Geometry box AI MATCH CONFIDENCE 53.1% POTENTIAL MATCH
No additional details provided.

Figure 4: AI Match Suggestions Modal — Illustrates the 4-factor hybrid matching engine displaying percentage match scores, visual tag overlap badges, and 1-click claim launcher.

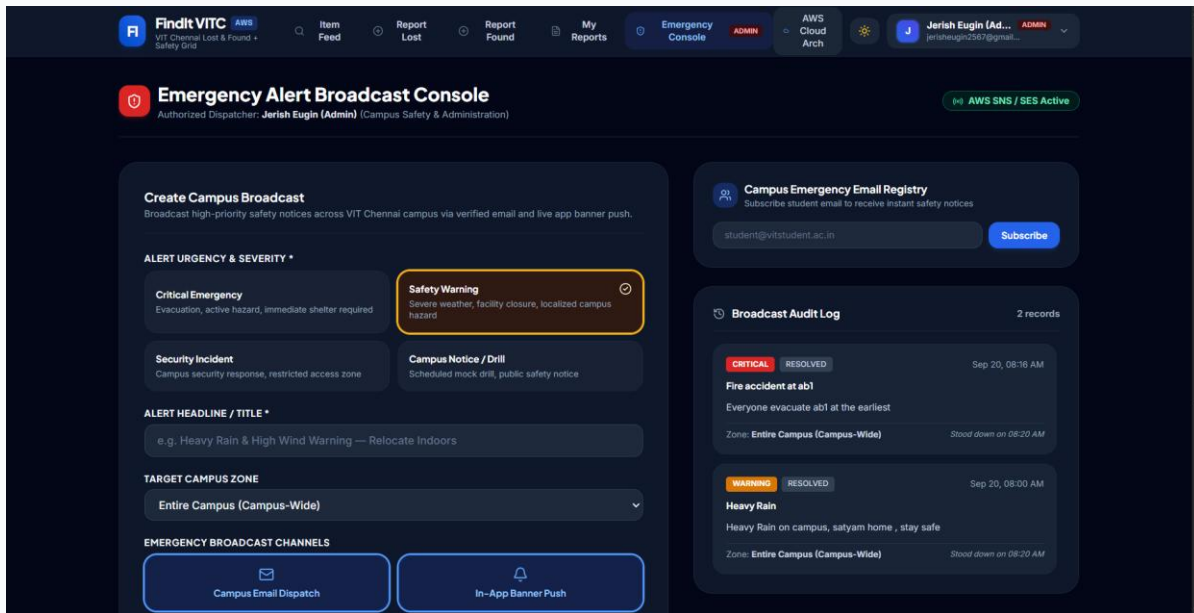


Figure 5: Emergency Alert Broadcast Console & Stand-Down Controls — Shows the admin emergency composer with severity selectors, target zones, SES email dispatch, live audit log, and 1-click alert termination.

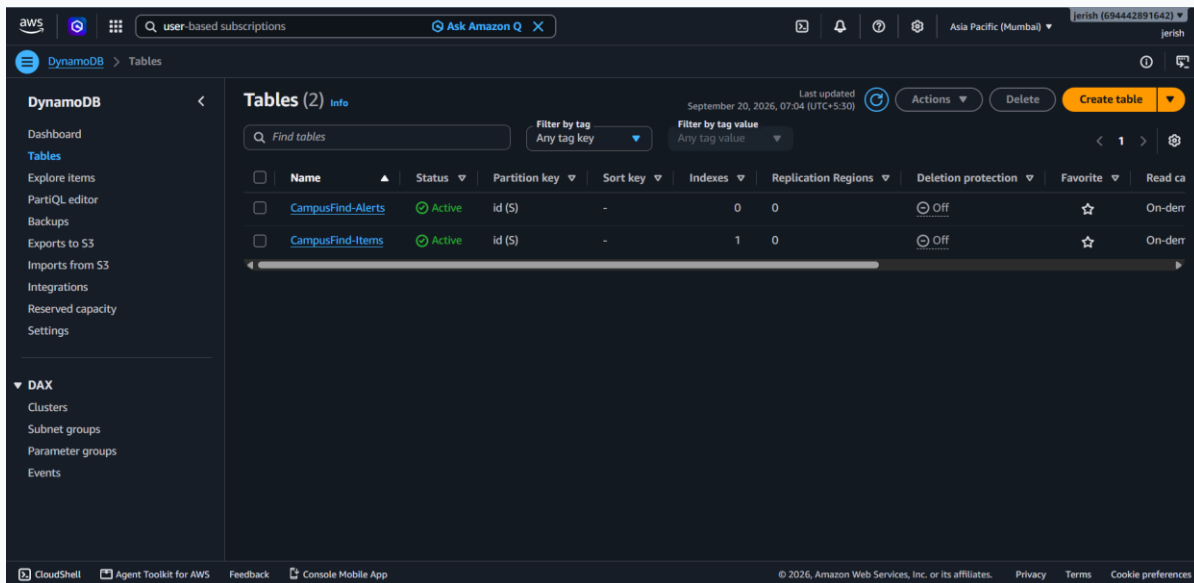


Figure 6: AWS DynamoDB Console — FindItVITC Tables & GSI Index — AWS ap-south-1 console view showing FindItVITC-Items table with TypeCreatedAtIndex Global Secondary Index and FindItVITC-Alerts table.

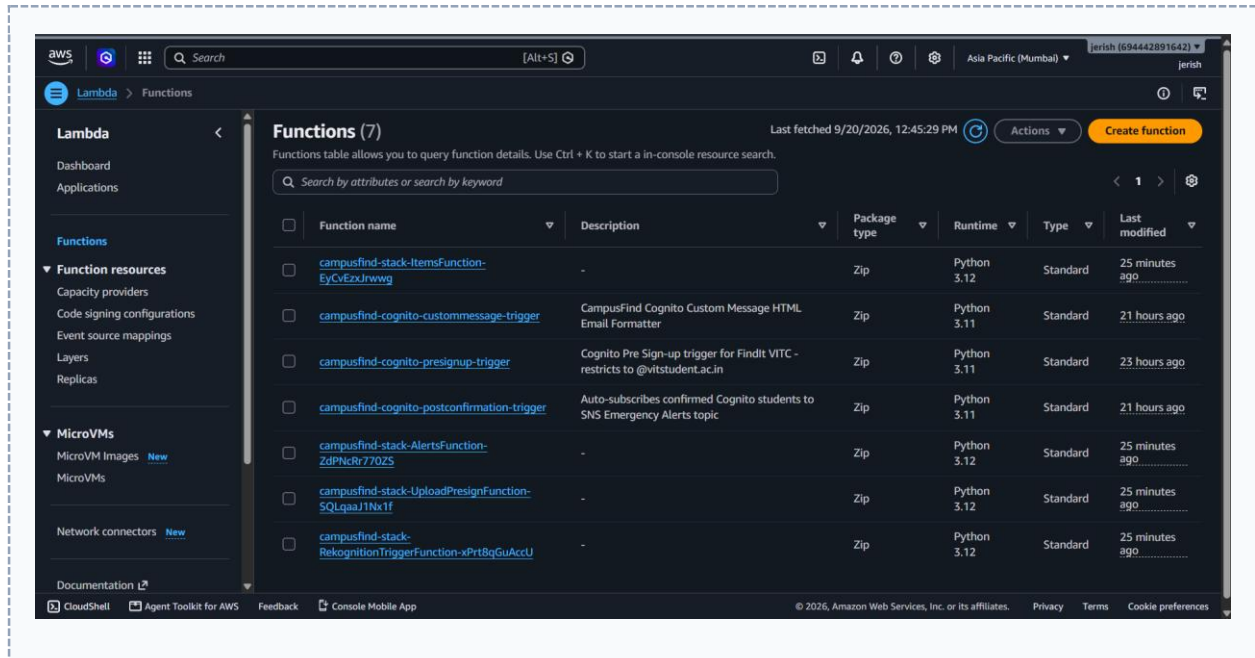


Figure 7: AWS Lambda Console — Deployed Serverless Microfunctions — AWS ap-south-1 console showing the four deployed microfunctions: ItemsFunction, AlertsFunction, RekognitionTriggerFunction, and UploadPresignFunction.

4.2 Representative Code Snippets & Analysis

Code Listing: Cognito Pre Sign-Up Domain Restriction Lambda (pre_signup.py)

```
def lambda_handler(event, context):
    trigger_source = event.get('triggerSource', '')

    # Allow Admin created users (campus safety staff)
    if trigger_source == 'PreSignUp_AdminCreateUser':
        event.setdefault('response', {})
        event['response']['autoConfirmUser'] = True
        event['response']['autoVerifyEmail'] = True
        return event

    user_attrs = event.get('request', {}).get('userAttributes', {})
    email = user_attrs.get('email', '').strip().lower()

    # Strictly enforce @vitstudent.ac.in domain restriction
    if not email.endswith('@vitstudent.ac.in'):
        raise Exception("Access Denied: Registration is strictly restricted to valid VIT Chennai
student accounts (@vitstudent.ac.in).")

    # Require 6-digit verification code dispatch
    event.setdefault('response', {})
    event['response']['autoConfirmUser'] = False
    event['response']['autoVerifyEmail'] = False
    return event
```

Snippet Analysis: This Lambda trigger executes synchronously before Amazon Cognito creates any user account. It inspects the submitted email attribute and raises an exception if the email does not end with '@vitstudent.ac.in'. This enforces security boundaries at the identity layer, preventing external unauthorized registrations while ensuring compliant onboarding for all VIT Chennai students.

Code Listing: Amazon Rekognition Computer Vision Processor (rekognition_processor.py)

```
def analyze_image_bytes(image_bytes: bytes, filename_hint: str = "", category_hint: str = "") -> Dict[str, Any]:
    norm_bytes, pixel_colors = normalize_image_and_extract_colors(image_bytes)
    client = get_rekognition_client()
    raw_labels, rek_colors = [], []

    if client:
        for min_conf in (45.0, 30.0):
            try:
                response = client.detect_labels(
                    Image={'Bytes': norm_bytes},
                    MaxLabels=15,
                    MinConfidence=min_conf,
                    Features=['GENERAL_LABELS', 'IMAGE_PROPERTIES']
                )
                labels = response.get('Labels', [])
                props = response.get('ImageProperties', {})
                # Extract dominant colors from foreground and background
                for c in (props.get('Foreground', {}).get('DominantColors', []) +
                        props.get('DominantColors', [])):
                    col = c.get('SimplifiedColor') or c.get('CSSColor')
                    if col: rek_colors.append(col.capitalize())
                for l in labels:
                    raw_labels.append({'name': l['Name'], 'confidence':
                        round(float(l['Confidence']), 1)})
                if labels: break
            except Exception as e:
                logger.warning(f"Rekognition evaluation error: {e}")

    # Fallback to semantic context if vision returns no labels
    if not raw_labels:
        raw_labels = extract_semantic_vision_tags(filename_hint,
            category=category_hint).get('detected_labels', [])

    return assemble_top_tags(raw_labels, pixel_colors + rek_colors)
```

Snippet Analysis: This function executes adaptive multi-confidence label detection (45% -> 30%) with ImageProperties color extraction. It normalizes arbitrary image formats (PNG, WEBP, JPEG), composites alpha channels onto clean backgrounds, and extracts top visual object labels and dominant colors. If the image contains no recognizable objects, it gracefully falls back to semantic taxonomy extraction.

Code Listing: DynamoDB Item Ingestion & Decimal Serialization (items.py)

```

def handle_create_item(body_data: Dict[str, Any], event: Dict[str, Any], user: Dict[str, Any],
table) -> Dict[str, Any]:
    item_id = str(body_data.get('id') or uuid.uuid4())
    now_iso = datetime.now(timezone.utc).isoformat()

    # Prevent future date spoofing
    raw_date = body_data.get('dateTime') or now_iso
    validate_datetime_not_future(raw_date)

    # Assign verified identity from Cognito token claims
    creator_user_id = user['userId'] if user['userId'] != 'anonymous-user' else
body_data.get('userId')
    creator_user_email = user['email'] or body_data.get('userEmail', '')

    new_item = {
        'id': item_id,
        'title': str(body_data['title']).strip()[:150],
        'type': str(body_data['type']).lower().strip(), # Partition key for GSI
        'category': str(body_data['category']).strip(),
        'location': str(body_data['location']).strip(),
        'dateTime': raw_date,
        'description': str(body_data.get('description', '')).strip()[:1000],
        'photoUrl': body_data.get('photoUrl', ''),
        'ai_tags': body_data.get('ai_tags', []),
        'detected_labels': convert_floats_to_decimals(body_data.get('detected_labels', [])),
        'status': 'open',
        'userId': creator_user_id,
        'userEmail': creator_user_email,
        'createdAt': now_iso, # Sort key for GSI
        'updatedAt': now_iso
    }

    table.put_item(Item=new_item)
    return build_cors_response(201, {'message': 'Item created successfully', 'item': new_item})

```

Snippet Analysis: This handler stores new lost/found reports in DynamoDB. It validates that incident timestamps cannot be in the future, binds verified Cognito user identities to prevent IDOR spoofing, and converts Python float metrics to Decimal types (preventing Boto3 serialization crashes). The record is indexed on the TypeCreatedAtIndex GSI for sub-second query retrieval.

Code Listing: API Gateway CORS & Cognito Authorizer SAM Specification (template.yaml)

```

FindItVITCApi:
  Type: AWS::Serverless::Api
  Properties:
    StageName: prod
    GatewayResponses:
      DEFAULT_4XX:
        ResponseParameters:
          Headers:
            Access-Control-Allow-Origin: "*"

```

```

        Access-Control-Allow-Headers: "'Content-Type,Authorization,X-Amz-Date,X-API-Key'"
        Access-Control-Allow-Methods: "'GET,POST,PATCH,PUT,DELETE,OPTIONS'"
    DEFAULT_5XX:
        ResponseParameters:
            Headers:
                Access-Control-Allow-Origin: "'*'"
                Access-Control-Allow-Headers: "'Content-Type,Authorization,X-Amz-Date,X-API-Key'"
                Access-Control-Allow-Methods: "'GET,POST,PATCH,PUT,DELETE,OPTIONS'"
    Auth:
        DefaultAuthorizer: CognitoAuth
        AddDefaultAuthorizerToCorsPreflight: false
        Authorizers:
            CognitoAuth:
                UserPoolArn: !GetAtt FindItVITCUserPool.Arn
                Identity:
                    Header: Authorization

```

Snippet Analysis: This CloudFormation/SAM configuration defines the REST API Gateway. Crucially, it attaches custom GatewayResponses (DEFAULT_4XX and DEFAULT_5XX) with explicit Access-Control-Allow-Origin headers. This prevents API Gateway from stripping CORS headers when Cognito authorizers reject invalid tokens, eliminating opaque client-side network errors.

5. Conclusion

5.1 Summary of Accomplishments

FindItVITC (FindItVITC) demonstrates how a complex, campus-wide operational workflow can be modernized using a cohesive cloud-native serverless architecture. By orchestrating Amazon Cognito, Amazon S3, Amazon Rekognition, AWS Lambda, Amazon DynamoDB, Amazon SNS, Amazon SES, and Amazon API Gateway into a unified event-driven system, the platform successfully eliminates the fragmentation and delays inherent in traditional physical lost-and-found operations.

Key engineering achievements include:

1. Complete Serverless Implementation: Achieved 100% serverless infrastructure with \$0.00 idle hosting cost, fully eligible under the AWS Free Tier.
2. AI Computer Vision Integration: Automated item categorization and dominant color tagging via Amazon Rekognition, reducing descriptive friction for students.
3. Multi-Factor AI Matching: Developed a 4-factor weighted similarity engine ($\geq 75\%$ confidence threshold) that proactively identifies candidate item matches.
4. Real-Time Emergency Grid: Built an administrative broadcast and stand-down subsystem delivering multi-channel alerts via Amazon SES email and synchronized in-app banners.
5. Production Deployment: Successfully deployed live on AWS ap-south-1 with a responsive, dark/light mode React frontend hosted at <https://finditvitc.github.io>.

5.2 Key Architectural Learnings

Throughout the design and implementation of FindItVITC, several valuable cloud engineering lessons were mastered:

- Direct-to-S3 Presigned Uploads: Offloading binary file uploads directly to S3 via presigned PUT URLs proved essential to prevent Lambda RAM exhaustion and avoid API Gateway payload limits.
- DynamoDB Type Serialization: Managing Python float conversions to Decimal types is mandatory when storing Rekognition confidence scores and numerical metrics in DynamoDB via Boto3.
- API Gateway GatewayResponses: Overriding default 4xx and 5xx API Gateway responses with explicit CORS headers is critical to ensure authorizer failures return clean JSON rather than browser CORS blocks.
- Client-Side Resilience: Implementing multi-tier fallbacks (client canvas compression, resilient local stores) ensures the application remains responsive even under intermittent network connectivity.

5.3 Limitations & Future Scope

While FindItVITC achieves all core design objectives, several promising avenues for future enhancement exist:

- Optical Character Recognition (OCR): Enhancing Rekognition pipelines with Amazon Textract to automatically read student registration numbers from photos of lost VIT identity cards and textbooks.
- Web Push & Progressive Web App (PWA): Integrating Web Push API and Service Workers to deliver native lock-screen emergency notifications directly to student mobile devices.

- Secure Physical Handover Lockers: Integrating smart IoT campus lockers where found items can be deposited, generating one-time QR codes for verified student retrieval.
- Native Mobile Applications: Compiling cross-platform iOS and Android mobile binaries using React Native for tighter operating system integration.

6. References

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